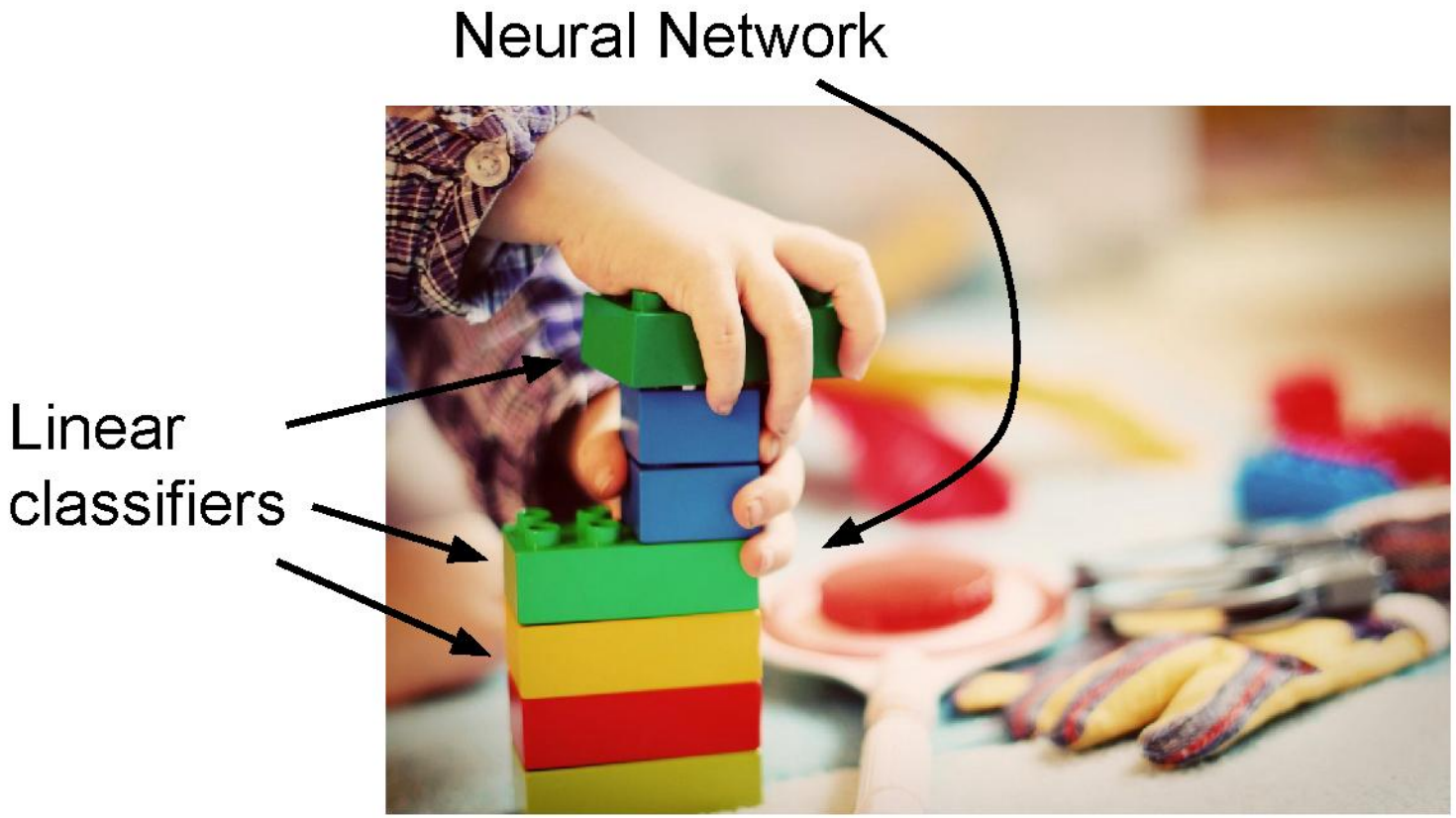
Classifiers

- Nearest Neighbor
- kNN (“k-Nearest Neighbors”)
- **Linear Classifier**
- **Neural Network**
- Deep Neural Network
- Transformers
- …

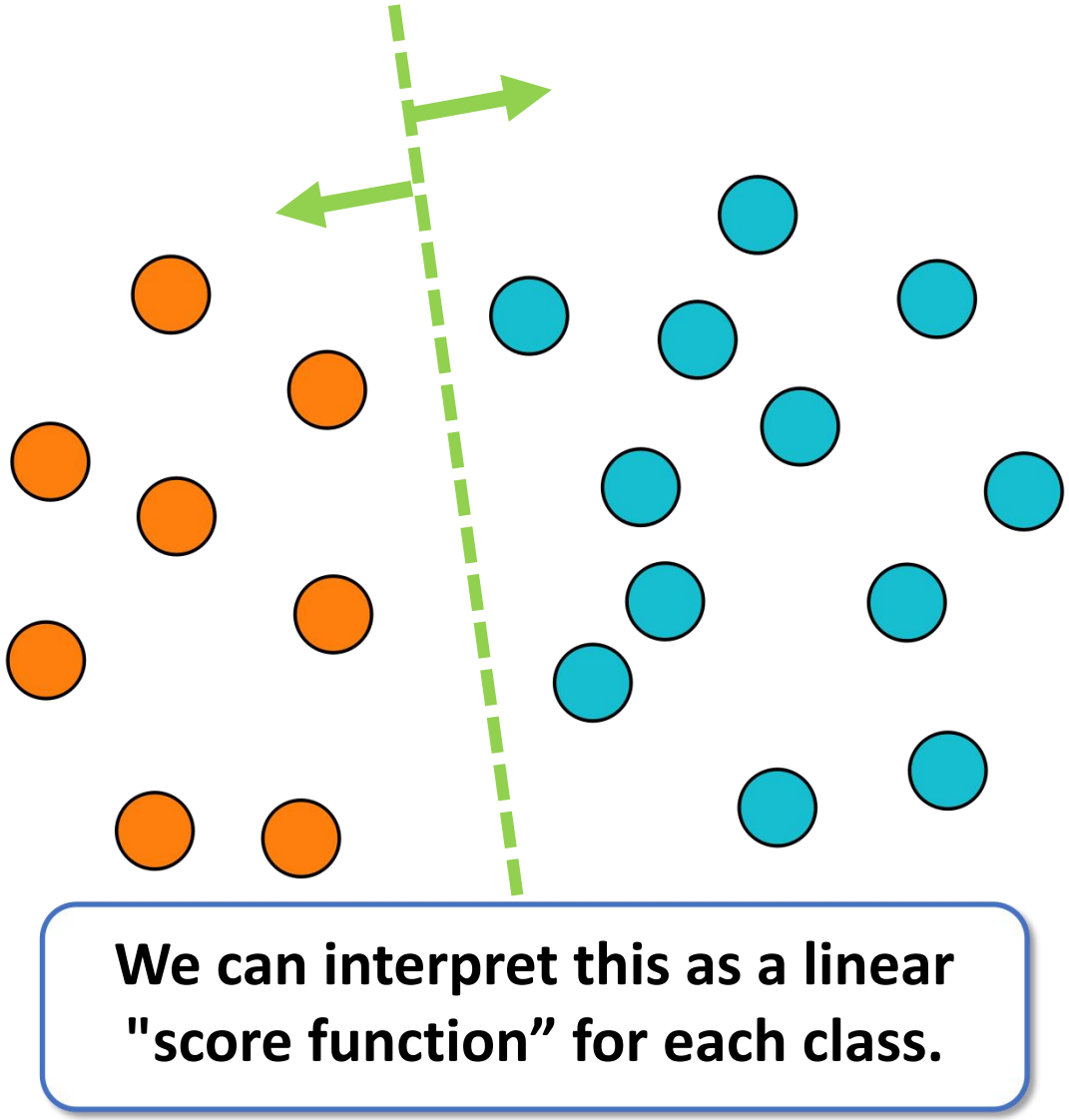
Linear Classifiers



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Linear Classification vs. Nearest Neighbors

- Linear Classifier
  - Store hyperplanes that best separate different classes
  - We can compute continuous class score by calculating (signed) distance from hyperplane



Score functions





class scores

Parametric Approach



image parameters

f(**x**,**W**)

10 numbers, indicating class scores

[32x32x3] $= 3072$
array of numbers 0...1
(3072 numbers total)

Parametric Approach: Linear Classifier

$$f(x, W) = Wx$$

3072x1

10x1

10x3072



10 numbers, indicating class scores

[32x32x3] $= 3072$
array of numbers 0...1

parameters, or “weights”

Slide adapted from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/

Parametric Approach: Linear Classifier

$f(x, W) = Wx$ + (+b)

3072x1

10x1

10x3072

10x1



10 numbers, indicating class scores

[32x32x3] $= 3072$
array of numbers 0...1

parameters, or “weights”

Slide adapted from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/

Linear Classifier

define a **score function**

data (image)

$$f(x_i, W, b) = Wx_i + b$$

class scores

"weights"

"bias vector"

"parameters"

Slide adapted from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/